

Chapter 1

Improving Classroom Quality and Children's Learning in Early Childhood Education: Evidence From a Policy–Research Partnership in Ghana

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Abstract

This chapter critically reviews opportunities and challenges in designing and conducting actionable research on improving preprimary school quality and children's learning and development, with a case study in Ghana. We describe the development of a research–practice partnership among university-based researchers, a local research organization, and Ghana's Ministry of Education. The partnership developed and evaluated an in-service teacher training and a parental awareness program. Both programs pursued the same goals of increasing child-centered and play-based learning and improving children's school readiness. We discuss the successes and challenges of scaling-up a successful teacher in-service training program. In addition, we describe unexpected findings of the parental awareness program that pose several new research questions. Implications for future research on measuring and improving pre-primary educational quality through similar research–practice partnerships are discussed.

Keywords: Early childhood education; preprimary; classroom quality; teacher training; school readiness; Ghana; sub-Saharan Africa

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Children's participation in early childhood education (ECE) is increasing worldwide. Many high-income countries have already achieved nearly universal ECE access for children ages three to five (OECD, 2015). ECE access is lower in low- and middle-income countries (LMICs), with regional averages ranging from 17.9% of 3- and 4-year-olds in sub-Saharan Africa (SSA) to 61.7% in Latin America and the Caribbean (McCoy et al., 2018). Despite relatively low enrollment in SSA, demand for ECE is growing. Research on ECE quality in SSA is limited. Process quality – which refers to the nature of children's daily interactions and experiences in the classroom – is considered the driver of child development and learning (Pianta, 2005). Structural quality – which refers to regulable resources, such as class size, student–teacher ratio, and teacher training and education (Slot et al., 2015) – is considered important for improving learning outcomes only to the extent that it promotes process quality (Phillips et al., 2000; Pianta, 2005). Global reports suggest that overall, the structural quality of services in SSA is low (UNESCO, 2015), while research on process quality and children's learning is scarce. For governments to successfully monitor and improve ECE quality at scale, researcher–governmental partnerships are needed to help define context-specific quality and develop both monitoring tools and teacher training models.

This chapter summarizes a 5-year partnership among US.-based academics, Ghana's Ministry of Education (MOE), a local research organization, and other stakeholders to develop and evaluate a nationally scalable in-service teacher training and parental awareness program to improve ECE process quality and children's learning. We reflect on what it takes to grow and sustain a multi-stakeholder partnership. We also consider the role of a critical partner based in Ghana – Innovations for Poverty Action (IPA) – and the long-term and ongoing relationship building they conducted with the government that was central for the project. We focus on key areas in assessing research and policy partnerships as proposed by Georgalakis and Rose (2019), including bounded mutuality, sustained interactivity, and policy adaptability, and consider each of these qualities in reflecting on the processes and outcomes related to the project.

Ghana's ECE Landscape

In 2004, Ghana's government adopted the National Early Childhood Care and Development (NECCD) Policy, which highlighted access to quality ECE as central to improving early childhood development and reducing learning inequalities. In 2007, 2 years of ECE – called *kindergarten 1* (KG1; for 4-year-olds) and *kindergarten 2* (KG2; for 5-year-olds) – were added to the universal basic educational system that previously had begun in primary school. As a result, Ghana has one of the highest ECE enrollment rates in SSA at 75% net enrollment in 2015–2016 (Ghana Ministry of Education, 2016). But classroom quality and learning are low (Ghana Education Service, 2013).

Similar to primary schools, Ghanaian ECE tends to focus primarily on academic skill development (Agbenyega, 2018), reflecting the demands of many parents for highly structured and “academically rigorous” education (Bidwell & Watine, 2014; Jukes et al., 2018). Researchers have raised concerns that the “traditional” educational approaches used in many African countries (e.g., rote instruction and memorization) are developmentally inappropriate for young children (Agbenyega, 2018; Akyeampong, 2017). These practices contrast with “modernist” approaches to ECE (i.e., child-centered, activity- and play-based approaches) that address social skills and unique needs of very young children (Hirsh-Pasek et al., 2009). When adopting the NECCD Policy, the government also created a kindergarten curriculum rooted in progressive, modernist approaches. The curriculum stipulates that play and creative activities, as opposed to academic work, should be central (Ghana Education Service, 2013). Both public and private ECE programs are expected to comply with the national curriculum. Yet a 2012 Government Kindergarten Situational Report concluded that teachers had not been systematically trained in the curriculum and that most teachers had not incorporated the curriculum into their practices (Ghana Education Service, 2012). The report concluded that training kindergarten teachers was a top educational policy priority, with a secondary priority to engage parents.

Public and private sectors. The private ECE sector in Ghana, as elsewhere in SSA, has grown recently. The sector has responded quickly to families’ demands for expanded ECE, particularly in urbanizing areas. The majority of these private ECE programs charge low fees, cater to low-income families, and struggle to cover their costs (Baum et al., 2018). Parents often perceive them to be of better quality than public programs (Dixon & Tooley, 2012; Zuilkowski et al., 2018), contributing to their growth (e.g., Bidwell & Watine, 2014).

While public and private kindergartens in Ghana are required to follow the national curriculum, a major concern for policy makers remains low educational quality and inadequate standards enforcement in both sectors. However, the public and private sectors differ in terms of structural characteristics. For example, there are few requirements for private sector teacher credentials, while public sector teachers are required to have diplomas in Basic Education obtained from approved colleges of education (Asare & Nti, 2014). Public sector teachers are civil servants and receive guaranteed remuneration levels and job security, while private sector teachers are generally paid less with no job security (Osei, 2006).

The role of parents. Parenting is a critical input to early childhood human capital formation (e.g., Schoellman, 2016). In our early meetings, stakeholders raised the important roles of parents in ECE and advocated that parents be included in the project. In promoting achievement across primary and secondary school, significant roles of families, family-school relations, and parental involvement in education have been clearly identified (e.g., Hill & Chao, 2009). Whether the same associations for parental involvement hold in low-resourced, low-literacy environments like Ghana, where many parents themselves have not completed their primary or secondary education, is much less researched.

While parenting practices in early childhood have been well-documented and researched across diverse contexts (e.g., Bornstein et al., 2015; Mesman et al., 2018), less research exists on parents' ECE involvement and their engagement with teachers in SSA. For younger children, regular communication between parents and schools may be particularly beneficial as it allows for working together toward school readiness. At the time of our study, there was very little literature on parental engagement in ECE settings and its effects on children, or on evidence-based intervention strategies to promote parental engagement.

The Quality Preschool for Ghana Project (QP4G)

The QP4G project began in 2014. IPA had completed a scoping study of the state of ECE in Accra (along with three other African cities) and was in discussions with Ghana's MOE about next steps. A MOE priority was to improve ECE quality. IPA reached out to the authors to collaborate to find or develop a cost-effective scalable model of transformational teacher training and to test the added benefits of engaging parents via an awareness campaign to align parental expectations with these new teacher practices.

At project initiation, a steering committee composed of various stakeholders was established that met twice in 2015 to guide the project, offer advice and direction, and provide insight into how best to address some of the pitfalls that would arise over the course of the project. The Steering Committee was responsible for ensuring the feasibility of the QP4G project in a direction consistent with the overall plans of the MOE. Steering Committee members participated in field visits to the project's schools. In addition to actively engaging the steering committee, a Policy Level Influence Plan was created to strategically engage decision makers to ensure the study remained a priority for policymakers and private sector influencers. This was done through consistent communication, holding meetings with key stakeholders, and incorporating various influencers into different parts of the implementation through monitoring, coaching and general oversight of trainings.

At the initial planning meeting, we discussed the needs of ECE teachers and the importance of conducting an impact evaluation to understand causal program effects. The project that was devised in collaboration with the advisory group aimed to build capacity and support for implementation of the 2004 kindergarten curriculum and to enhance the quality of kindergarten education. The goal was to develop and rigorously evaluate a scalable model of transformational teacher training to provide high-quality ECE and to test the benefits of a parental awareness campaign designed to align parental expectations with these practices. The programs were designed to improve classroom process quality and to enhance the development of children's school readiness within existing educational systems (i.e., district governments, a local teacher training center, and schools).

The implementing organization for the teacher program was the National Nursery Teacher Training Center (NNTTC), a teacher training center that is

funded partially by the Ghana Education Service (GES) in the MOE. NNTTC is the primary training center for ECE certification in Accra. We worked with NNTTC leadership and a nonprofit organization, Sabre Education, to condense and enhance their 8-week preservice training to be delivered in a 5-day in-service training format and to train district ECE coordinators to serve as teacher coaches. The parental awareness intervention (three videos) was created by hired local media consultants and implemented by ECE coordinators who work in the district education offices, who screened the videos and led discussions with caregivers through school Parent–Teacher Association meetings. IPA trained these district officers, and IPA and the researchers oversaw the video creation.

Six of the nine most disadvantaged districts in the Greater Accra Region were selected based on distance from the teacher training facility. Two-hundred-and-forty schools were randomly sampled, stratified by (a) district, and (b) public/private sector. Programs were randomized after the first round of baseline data collection to one of three conditions: (1) in-service teacher training program only; (2) in-service teacher training program plus parental awareness program; or (3) control condition.

Impact Evaluation Results

The impact evaluation allowed the project partners and stakeholders to test the two programs' effectiveness in improving teaching quality and child outcomes. Our regular engagement with our steering committee and governmental partners helped establish program ownership and led to vested interests in the research findings. We found that this brief, affordable, and scalable program improved several dimensions of teaching quality, teacher knowledge about early childhood development, and children's school readiness. The teacher training and coaching program produced a replicable and scalable model for the government to improve ECE quality in peri-urban areas (Wolf et al., 2019). Impacts on classroom processes (including Emotional Support and Behavior Management, and Supporting Student Expression) were moderate – 0.5–0.6 standard deviations – while impacts on children's academic and social–emotional outcomes were small – 0.13–0.18 standard deviations. These effect sizes are in line with many ECE teacher professional development studies. We found little evidence of impact heterogeneity by gender or baseline proficiency, suggesting that QP4G improved outcomes for these subgroups equally.

We also found moderate-sized reductions, much larger in the private sector, in teacher burnout and leaving the profession mid-year. Private sector teachers had lower educational qualifications and less teaching experience and were significantly younger compared with public sector teachers. The training may have improved self-efficacy among private sector teachers who had relatively fewer skills and training at baseline than public sector teachers. One implication is that private sector teacher retention may improve if schools invest in effective teacher professional development.

Results of the added effects of the parental awareness intervention were surprising. Rather than enhance teacher training program effectiveness, the parental awareness intervention *counter-acted* impacts on some dimensions of classroom quality and on all child outcomes. This led to follow-up qualitative research in which we found that parents worried that child-centered and play-based learning structures would hinder their children's learning and pushed back on teachers implementing these methods. In addition, parents viewed their roles as providing basic necessities for their children, rather than “teachers” at home, and relied on teachers to support their children's learning (Wolf, 2020).

Importantly, when the originally planned study ended, we raised further grant funds to continue to follow the children, making it the first longitudinal impact evaluation of an ECE quality improvement intervention in SSA. These additional follow-ups have allowed us to examine medium-term persistence and fade-out of impacts. We found small persistent impacts of the teacher training intervention for children's social-emotional skills 1 year later (Wolf, Aber, Behrman, & Peele, 2019), and persistent impacts for children's language and executive function skills 2 years later (Wolf, 2019). We also found negative impacts of the parental awareness program on numeracy outcomes 2 years later (Wolf, 2019).

Policy Engagement

The impact evaluation led to important evidence-based insights on the two programs' effectiveness. We leveraged lessons from the implementation of and evidence from the project toward the achievement of three broad goals: (a) policy engagement, development, and implementation, (b) capacity strengthening, and (c) dissemination and data use. These activities have helped educationalists and policymakers consider how the QP4G findings could be used in their work through program planning, budgets, educational programs, or policy formulation and implementation.

Policy engagement, development, and implementation. In line with our team's commitment to cocreating evidence and empowering policy makers to use evidence, we undertook several relationship-based engagement activities: stakeholder engagement, supporting ECE policy development, and supporting educational programming and implementation. This sustained interactivity led to a broad base of support and investment in the project.

Our efforts to maintain stakeholder engagement throughout the project included a wide, participatory process with our steering committee of designing intervention tools and materials to ensure buy-in from all stakeholders and promote program sustainability. This included quarterly meetings with membership comprising IPA, GES, private school associations, UNICEF, and NNTTC. After the project, our IPA partners continued to support the development of Ghana's ECE policy, working with the Technical Working Group for ECE policy to make modifications, reviewed the final version of the indicator framework, and made additional ECE policy inputs. The broad goal of the policy is to create a systematic accountability framework to strengthen the subsector for efficient and

effective kindergarten service delivery to improve equitable access and quality education for all 4- to 5-year-olds. The policy focuses on five areas, including planning and management, curriculum development and implementation, teacher pre- and in-service training, family and community engagement, and monitoring, regulation, and quality assurance. The lessons from QP4G implementation and results provided considerable evidence to support the inputs that IPA made into the policy. New ECE policy was presented to the cabinet in August 2020, approved in September 2020, and launched in 2021.

Finally, our efforts to support educational programming and implementation were guided by the QP4G findings, which helped to shape the content creation and implementation of key government-led educational programs. IPA provided input into the National Teaching Council's (NTC) compliance framework for ECE service providers, which sets guidelines for ECE teacher training institutions (both preservice and in-service) and ECE instructors or teachers. The content includes play-based teacher training approaches, mentoring and coaching for teachers post-training, and equipping teachers with skills to engage parents/communities. The training module of this compliance framework is expected to be aligned with the ECE policy: Ghana Accountability for Learning Outcomes Project (GALOP). IPA contributed toward the development of the training and school-based support and instructional leadership modules for GALOP, using lessons learned and evidence from the QP4G project. In addition, IPA supported the creation of assessment forms for the new kindergarten curriculum based on the QP4G project for the National Council for Curriculum and Assessment. These forms are expected to guide teachers' timely assessments of children's progress in kindergarten. In addition, IPA participated in workshops to develop the tools and also reviewed tools prepared for all four competency areas of the curriculum.

After QP4G ended, IPA also supported the creation of kindergarten inspection tools for the National Inspectorate Board through review and inputs on initial drafts. The kindergarten inspection tools cover lesson observation, school leadership, and compliance with MOE requirements for the establishment of a kindergarten. The culmination of this work is a plan by IPA and Pre-Tertiary Directorate of MOE to create an ECE Evidence Committee to ensure the alignment of evidence from ECE research with MOE and GES initiatives and to strengthen policy implementation. The ECE Evidence Committee is expected to be modeled after the QP4G Steering Committee but broadened to cover all ECE-related research.

Finally, we also conducted a comprehensive costing analysis of the teacher training and coaching program to inform our governmental partners of the financial costs for any future scale-up. We found that the QP4G per-teacher cost was \$402, which translates into a per-child cost of \$16. These estimates provide concrete figures for the government to consider when scaling-up the program.

Capacity strengthening. The QP4G project had an inherent capacity-strengthening strategy targeted at the district educational officials who participated in the project. The project was designed to use existing systems and structures at the district and national educational offices to build district

educational officials' capacities and ensure scalability. The capacity strengthening goal was to narrow the capacity gaps of district educational officials by equipping them with the necessary tools and resources and enhancing their skills and knowledge for conducting monitoring and coaching for teachers and head teachers and facilitating parental awareness trainings.

Dissemination and data use. Part of IPA's mission is to disseminate research results and facilitate their use in program design and evaluation through participatory conferences, print materials, and web-based tools that translate data into publicly accessible information. The academic researchers and IPA representatives gave presentations at both national and international conferences including at the Ghana National Education Week (2017, 2018, 2019), Accra; Ghana Teacher Prize (2018), Accra; Public Finance and Public Management in Africa workshop (2018), Accra; and Comparative and International Education Society (CIES 2019) in California, USA. We also created print materials, including policy briefs, academic publications, and project evaluation reports. Finally, the project summary, teacher training and survey materials, academic publications, and research results were posted on the project website and linked to deidentified survey datasets published on the World Bank Microdata Library.^{1,2} Varied users including implementing organizations, academics, development practitioners, and educationists across the world have accessed and used these materials.

Challenges and New Directions

We encountered several limitations and challenges throughout the project, some of which have led to new directions for our research and partnerships. The first and perhaps most important challenge pertains to the surprising counter-active impacts of the parental awareness intervention on teaching and child learning outcomes. It is clear that a better understanding of parental roles in shaping educational quality in SSA is needed. The counter-intuitive results were surprising but uncovered important dynamics between parents and teachers that have not been considered to date and that likely will continue to shape educational reform efforts in many countries. Interestingly, a recent randomized control trial (RCT) in Mexican secondary schools found similar results, where student achievement at schools in low-income areas that provided parents with information on supporting their children's education had no impacts. In an additional treatment arm where parents also received extra money and some decision-making power, student achievement also did not improve, and teacher trust eroded (Barrera-Orsorio et al., 2020). These findings, along with ours, suggest that a greater basic understanding of parent-teacher relationships is needed in order to strengthen – rather than weaken – them, in future intervention research.

¹<http://www.poverty-action.org/study/improving-kindergarten-quality-ghana>

²<https://datacatalog.worldbank.org/dataset/ghana-quality-preschool-impact-evaluation-2017>

Second, our longitudinal data collection with teachers one year later enabled us to assess sustained gains from the intervention year. Unfortunately, we found that most of the positive impacts on teaching quality faded out by the end of the next school year (Wolf & Peele, 2019). These results indicated that teachers likely need ongoing support after a program is administered, and that ongoing training is critical to sustain quality improvement efforts. This is a lesson relevant for teacher professional development programs more generally.

Third, we lacked comprehensive measures of fidelity of implementation, leading to a lack of understanding of the key components of the intervention needed for successful scale-up (e.g., refresher trainings, coaching, specific content modules from the training manual). This limited our ability to inform the government of the most important training components, as scaling-up costs require some program elements to be cut. Including measures of implementation fidelity in future RCTs is crucial to understand in which ways the intervention implementation is related to treatment impacts.

Fourth, despite success of the teacher training program, there is still limited scalability to date. While NNTTC agreed to offer the 5-day training program in partnership with Ghana Private School Association, this did not come to fruition. Another local NGO partner, Sabre Education, has raised private funds to provide the in-service training to specific districts in Ghana, but no systematic effort in partnership with the government or the private sector has materialized.

Fifth, we did not sufficiently consider the transition from preprimary to primary school. The transition to primary school represents a distinct adjustment for children. Research on school transitions considers them to be multi-year processes with each new setting or grade having unique demands for children, families, and teachers (Ramey & Ramey, 2010). At the same time, educators recognize, and families often experience, a lack of continuity between early childhood and early elementary school programs (e.g., La Paro et al., 2000), and this transition may be particularly important to children's longer term school adjustment. Children who can develop positive connections and foster communication with their teachers, peers, and families may have more successful adjustments in their new classrooms (Hamre & Pianta, 2001; Rimm-Kaufman & Pianta, 2000). Similar issues were raised by members of our Steering Committee, and yet this was not incorporated into our project. We later obtained funds to follow-up children in the first few years of primary school and found that children exposed to the teacher training intervention had better higher order cognitive functioning – measured as literacy and executive function outcomes – and this was particularly true in high-quality primary classrooms (Wolf, 2019). Moving forward, any efforts targeting ECE should also incorporate transitions to primary schools and harmonizing pedagogical approaches in the first years of primary school.

Sixth, we were restricted by logistical and budgetary considerations to working in the Greater Accra Region, to enable teachers to reach the NNTTC Accra training center. This limited our sample to urban and peri-urban communities, and thus we were not able to generalize the results to rural schools, where similar efforts are needed. It was clear that the interest in implementing QP4G in rural areas was great. But it is also clear that the program would need important

adaptations in order to work in rural contexts, including transportation challenges that limit local ECE coordinators from visiting teachers in their classrooms.

Finally, the COVID-19 pandemic brought education around the world to a halt. In Ghana, schools closed from March through December 2020. We were in the field collecting another round of data with the QP4G children in March 2020 and had to halt all activities. We have followed up with phone surveys with children, parents, and teachers to learn (a) how the pandemic has affected learning and development for the children and (b) whether participation in QP4G moderates the influence of the pandemic on children's learning and development. Nonetheless, our ability to track children has been hampered, partially limiting our understanding of the longer term consequences of the QP4G programs on children and their families.

Implications for Improving ECE Quality Through Policy–Research Partnerships

The QP4G project provides several important implications for future research on ECE and policy–research partnerships. First, our findings from additional follow-up data collection point to the necessity to conduct longitudinal studies on ECE interventions. The longer term persistence or fade-out of short-term positive gains from ECE has received attention in the United States (e.g. Bailey et al., 2017), but not yet in LMICs. A recent review of RCTs in LMICs found that very few studies have evaluated longer term impacts, and the authors highlighted a need for high-quality evidence on the long-run impacts of interventions in LMICs (Bouguen et al., 2019). Such longitudinal research is critical to inform the design of preprimary and primary school educational reforms and understand the broader implications for society.

Second, there are tensions between global versus localized conceptualizations and assessments of quality. On one hand, we wanted our measures to have international applicability and be in line with the broader literature on classroom quality. On the other hand, we wanted our measures to be tailored to the QP4G program and to align with the government's ECE goals. Indeed, measuring quality is a central element of scaling ECE programs within countries (Yoshikawa et al., 2018). The vast majority of existing classroom observation tools were developed in US contexts, and primarily used within these contexts. When we conducted our study, two efforts were underway to develop classroom observation tools for use in low-income country settings (MELE; Raikes et al., 2020; TIPPS; Seidman et al., 2018), though neither had been widely used or validated at the time. As part of our need for a classroom observation tool that was locally applicable, we contributed by conducting the first large-scale study testing the validity of the Teacher Instructional Processes and Practices Scale (TIPPS). After piloting the TIPPS and conducting cognitive interviews with teachers to assess its face validity in Ghana, we concluded that it was suitable for use in that context. We used the data collected during our baseline to test the psychometric properties

of the tool and examine the validity of the underlying factor structure by testing its associations with teacher characteristics and child outcomes, and published the first peer-reviewed article examining the validity of a tool designed for use in LMICs (Wolf et al., Behrman, 2018).

Third, it became clear throughout the course of our project that we must understand teachers' lives in context and how they affect (and are affected by) professional development. We found that teacher mental health and other contextual conditions affected both their take-up of the QP4G program and was affected by training. For example, job dissatisfaction predicted teacher attrition, and experiencing depressive symptoms was associated with lower take-up of teacher training (Schwartz et al., 2019). Personal poverty and time demands were associated with lower adherence to training materials and ultimately implementation quality (Schwartz et al., 2019). Further we found that a challenging workplace environment was associated with increased anxiety and depressive symptoms for teachers over the course of the school year. Interestingly, anxiety and depressive symptoms increased for all teachers over the school year. However, randomization to either intervention was linked to significantly smaller increases in symptoms over the school year (Peele & Wolf, 2020). Incorporating a more holistic view of teachers is needed to inform the successful design of professional development and to understand the full effects on teaching and teachers.

Fourth, we are exploring opportunities to conduct a re-randomization of the QP4G sample to an intervention in early adolescence. This would allow us to examine the additive and possibly multiplicative benefits of support to children during two critical developmental/school transitions: from early childhood to middle childhood (ECE to primary school) and from middle childhood to early adolescence (primary to junior secondary school). Such work could be built in to future ECE evaluations as well, with re-randomizing other critical intervention support during important transitions.

Finally, in reflecting on the successes and challenges of our partnership, it is clear that such policy–research partnerships require a large tent and a lead coordinating body (in our case, IPA Ghana) to continue to bring the research and evidence to the forefront of policy conversations and decisions. Such partnerships are also essential for tracking longer run program effects and integrating research results into scaled-up policies but require ongoing research support, which is challenging to obtain particularly since many funders are focused on very short-term results.

References

- Agbenyega, J. S. (2018). Examining early childhood education system in Ghana: How can Bourdieuan theorisation support a transformational approach to pedagogy? In M. Fleer & B. van Oers (Eds.), *International handbook of early childhood education* (Vol. 1, pp. 673–705). <https://doi.org/10.1007/978-94-024-0927-7>
- Akyeampong, K. (2017). Teacher educators' practice and vision of good teaching in teacher education reform context in Ghana. *Educational Researcher*, 46(4), 194–203. <https://doi.org/10.3102/0013189x17711907>

- Asare, K., & Nti, S. K. (2014). Teacher education in Ghana: A contemporary synopsis and matters arising. *Sage Open*, 4(2), 1–8. <https://doi.org/10.1177/2158244014529781>
- Bailey, D., Duncan, G. J., Odgers, C. L., & Yu, W. (2017). Persistence and fadeout in the impacts of child and adolescent interventions. *Journal of Research on Educational Effectiveness*, 10(1), 7–39. <https://doi.org/10.1080/19345747.2016.1232459>
- Barrera-Osorio, F., Gertler, P., Nakajima, N., & Patrinos, H. (2020). Promoting parental involvement in schools: Evidence from two randomized experiments. In *NBER working paper series (No. w28040)*. <http://www.nber.org/papers/w28040>
- Baum, D., Abdul-Hamid, H., & Wesley, H. (2018). Inequality of educational opportunity: The relationship between access, affordability, and quality of private schools in Lagos, Nigeria. *Oxford Review of Education*, 44(4), 459–475.
- Bidwell, K., & Watine, L. (2014). Exploring early education programs in peri-urban settings in Africa. *Optimus Impact*, 5, 3–19.
- Bornstein, M. H., Putnick, D. L., Bradley, R. H., Lansford, J. E., & Deater-Deckard, K. (2015). Pathways among caregiver education, household resources, and infant growth in 39 low- and middle-income countries. *Infancy*, 20(4), 353–376. <https://doi.org/10.1111/inf.12086>
- Bouguen, A., Huang, Y., Kremer, M., & Miguel, E. (2019). Using RCTs to estimate long-run impacts in development economics. *Annual Review of Economics*, 11, 523–561. <https://doi.org/10.1146/annurev-economics-080218-030333>
- Dixon, P., & Tooley, J. (2012). A case study of private schools in Kibera: An update. *Educational Management Administration & Leadership*, 40(6), 690–706.
- Georgalakis, J. & Rose, P. (2019). Introduction: Identifying the qualities of research-policy partnerships in international development – A new analytical framework. *IDS Bulletin Exploring Research–Policy Partnerships in International Development*, 50(1), 1–19.
- Ghana Education Service. (2012). *Programme to scale-up quality kindergarten education in Ghana*. Ghana Ministry of Education. https://issuu.com/sabretom/docs/10_12_12_final_version_of_narrative_op
- Ghana Education Service. (2013). *National plan for kindergarten education*. Ghana Education Service.
- Ghana Ministry of Education. (2016). *Education sector performance report, 2016*. https://sapghana.com/data/documents/Education+Sector+Performance+Report+ESPR+2015_Final.pdf
- Hamre, B., & Pianta, R. C. (2001). Early teacher-child relationships and the trajectory of children's school outcomes through eighth grade. *Child Development*, 72(2), 625–638.
- Hill, N. E., & Chao, R. K. (2009). *Families, schools, and the adolescent: Connecting research, policy, and practice*. Teachers College Press.
- Hirsh-Pasek, K., Golinkoff, R., Berk, L., & Singer, D. (2009). *A mandate for playful learning in preschool: Applying the scientific evidence*. Oxford University Press.
- Jukes, M. C. H., Gabrieli, P., Mgonda, N. L., Nsolezi, F., Jeremiah, G., Tibenda, J. L., & Bub, K. L. (2018). “Respect is an investment”: Community perceptions of social and emotional competencies in early childhood from Mtwara, Tanzania. *Global Education Review*, 5(2), 160–188.

- La Paro, K. M., Pianta, R., & Cox, M. (2000). Kindergarten teachers' reported use of kindergarten to first grade transition practices. *The Elementary School Journal*, 101(1), 63–78.
- McCoy, D. C., Salhi, C., Yoshikawa, H., Black, M., Britto, P., & Fink, G. (2018). Home- and center-based learning opportunities for preschoolers in low- and middle-income countries. *Children and Youth Services Review*, 88, 44–56. <https://doi.org/10.1016/j.chilcyouth.2018.02.021>
- Mesman, J., Minter, T., Angnged, A., Cissé, I. A. H., Salali, G. D., & Migliano, A. B. (2018). Universality without uniformity: A culturally inclusive approach to sensitive responsiveness in infant caregiving. *Child Development*. <https://doi.org/10.1111/cdev.12795>
- OECD. (2015). OECD family database. <http://www.oecd.org/els/family/database.htm>. Accessed on January 16, 2021.
- Osei, G. M. (2006). Teachers in Ghana: Issues of training, remuneration and effectiveness. *International Journal of Educational Development*, 26(1), 38–51. <https://doi.org/10.1016/j.ijedudev.2005.07.015>
- Peele, M., & Wolf, S. (2020, February). Predictors of anxiety and depressive symptoms among teachers in Ghana: Evidence from a randomized controlled trial. *Social Science and Medicine*, 253, 112957. <https://doi.org/10.1016/j.socscimed.2020.112957>
- Pianta, R. C. (2005). A new elementary school for American children. *SRCD Social Policy Report*, 19(3), 4–5.
- Raikes, A., Koziol, N., Davis, D., & Burton, A. (2020). Measuring quality of pre-primary education in sub-Saharan Africa: Evaluation of the measuring early learning environments scale. *Early Childhood Research Quarterly*, 53, 571–585. <https://doi.org/10.1016/j.ecresq.2020.06.001>
- Ramey, C. T., & Ramey, S. L. (2010). The transition to school: Concepts, practices, and needed research. In *Transitions for young children: Creating connections across early childhood systems* (pp. 19–32). Georgetown U, Ctr on Health and Education; Paul H Brookes Publishing.
- Rimm-Kaufman, S. E., & Pianta, R. C. (2000). An ecological perspective on the transition to kindergarten: A theoretical framework to guide empirical research. *Journal of Applied Developmental Psychology*, 21(5), 491–511. [https://doi.org/10.1016/S0193-3973\(00\)00051-4](https://doi.org/10.1016/S0193-3973(00)00051-4)
- Schoellman, B. T. (2016, July). Early childhood human capital and development author(s). *American Economic Journal: Macroeconomics*, 8(3), 145–174. <https://www.jstor.org/stable/24>
- Schwartz, K., Cappella, E., Aber, J. L., Scott, M. A., Wolf, S., & Behrman, J. R. (2019). Early childhood teachers' lives in context: Implications for professional development in under-resourced areas. *American Journal of Community Psychology*, 63(3–4), 270–285. <https://doi.org/10.1002/ajcp.12325>
- Seidman, E., Kim, S., Raza, M., Ishihara, M., & Halpin, P. F. (2018, April 1). Assessment of pedagogical practices and processes in low and middle income countries: Findings from secondary school classrooms in Uganda. *Teaching and Teacher Education*, 71, 283–296. <https://doi.org/10.1016/j.tate.2017.12.017>
- UNESCO. (2015). *Education for all global monitoring report regional overview: Sub-Saharan Africa*. UNESCO.

- Wolf, S. (2019). Year 3 follow-up of the 'Quality Preschool for Ghana' interventions on child development. *Developmental Psychology*. <https://doi.org/10.1037/dev0000843>
- Wolf, S. (2020). "Me I don't really discuss anything with them": Parent and teacher perceptions of early childhood education and parent-teacher relationships in Ghana. *International Journal of Educational Research*, 99, 101525. <https://doi.org/10.1016/j.ijer.2019.101525>
- Wolf, S., Aber, J. L., Behrman, J. R., & Peele, M. (2019, April). Longitudinal causal impacts of preschool teacher training on Ghanaian children's school readiness: Evidence for persistence and fade-out. *Developmental Science*, 1–19. <https://doi.org/10.1111/desc.12878>
- Wolf, S., Aber, J. L., Behrman, J. R., & Tsinigo, E. (2019). Experimental impacts of the "quality preschool for Ghana" interventions on teacher professional well-being, classroom quality, and children's school readiness. *Journal of Research on Educational Effectiveness*, 12(1), 10–37. <https://doi.org/10.1080/19345747.2018.1517199>
- Wolf, S., & Peele, M. E. (2019). Examining sustained impacts of two teacher professional development programs on professional well-being and classroom practices. *Teaching and Teacher Education*, 86. <https://doi.org/10.1016/j.tate.2019.07.003>
- Wolf, S., Raza, M., Kim, S., Aber, J. L., Behrman, J., & Seidman, E. (2018). Measuring and predicting process quality in Ghanaian pre-primary classrooms using the Teacher Instructional Practices and Processes System (TIPPS). *Early Childhood Research Quarterly*, 45, 18–30. <https://doi.org/10.1016/j.ecresq.2018.05.003>
- Yoshikawa, H., Wuerml, A. J., Raikes, A., Kim, S., & Kabay, S. B. (2018). Toward high-quality early childhood development programs and policies at national scale: Directions for research in global contexts. *Social Policy Report*, 31(1), 1–36.
- Zuilkowski, S., Piper, B., Ong'ele, S., & Kiminza, O. (2018). Parents, quality, and school choice: Why parents in Nairobi choose low-cost private schools over public schools in Kenya's free primary education era. *Oxford Review of Education*, 44(2), 258–274.

Author Query Form

Queries and/or remarks

[Q1]	The chapter title has been modified. Please check, and correct if necessary.
[Q2]	Missing reference: Ref "Slot et al., 2015; Phillips et al., 2000" are not listed in the "References" section; please provide complete reference details.